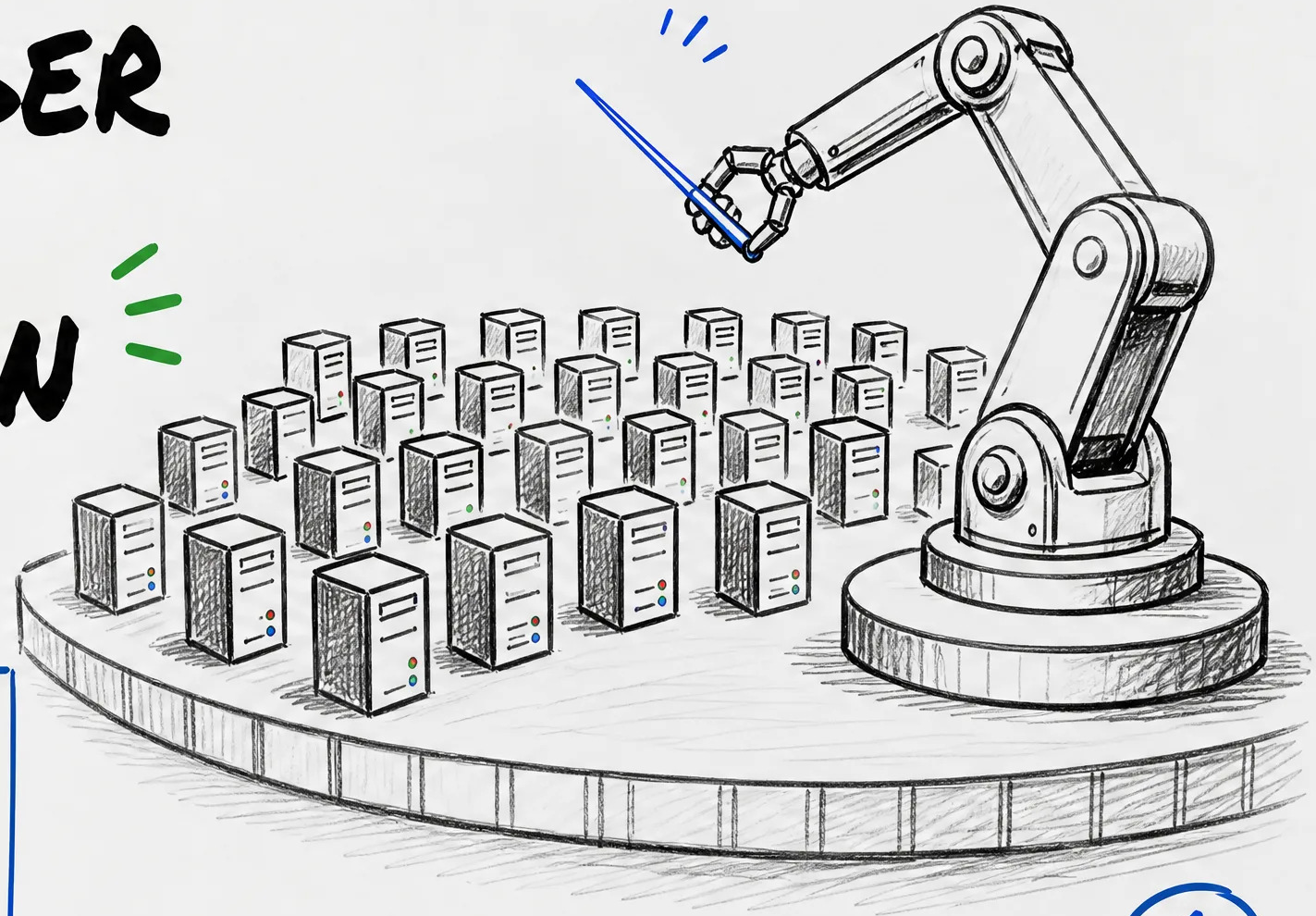
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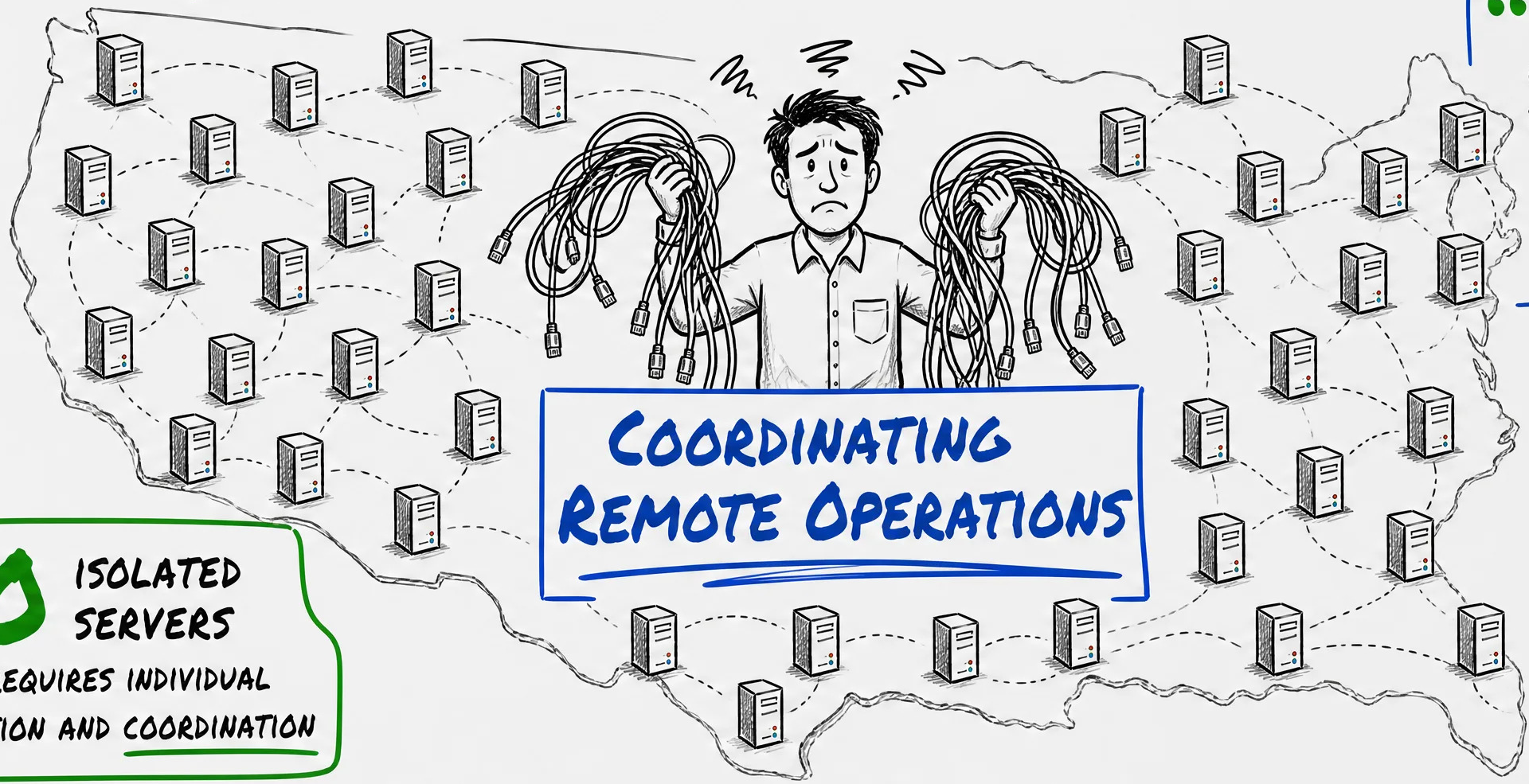
Job Manager and Automation



> Understanding a **deprecated** feature is valuable when it helps you operate an **existing estate** without designing a **new dependency** on it.

The Administrative Problem

Job Manager Solved



Coordinating Remote Operations

> "The more targets one action reaches, the more important targeting, timing, and verification become."

50 isolated servers

each requires individual attention and coordination

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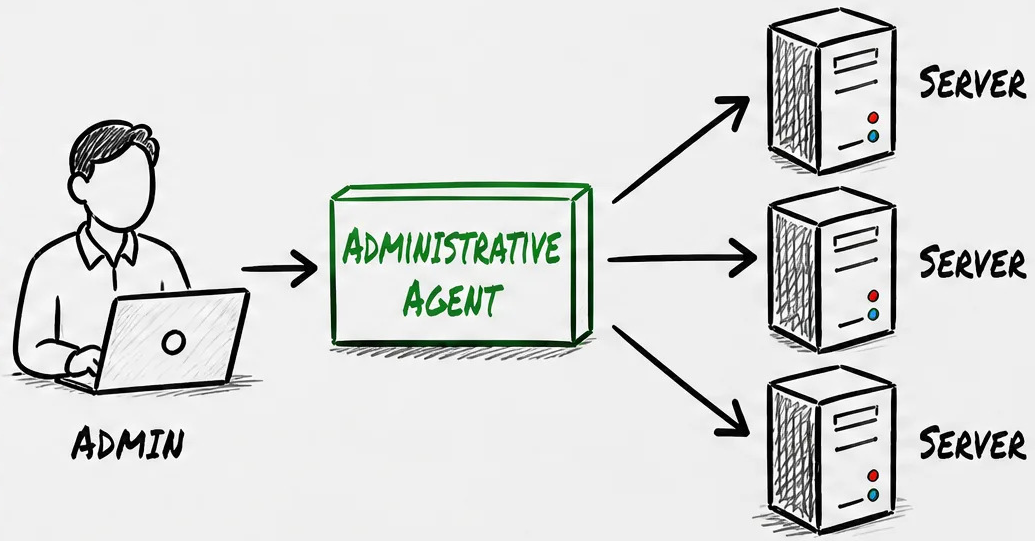
Independent Administration vs Administrative Agent

Independent Administration



Centralizing Local Administration

Administrative Agent



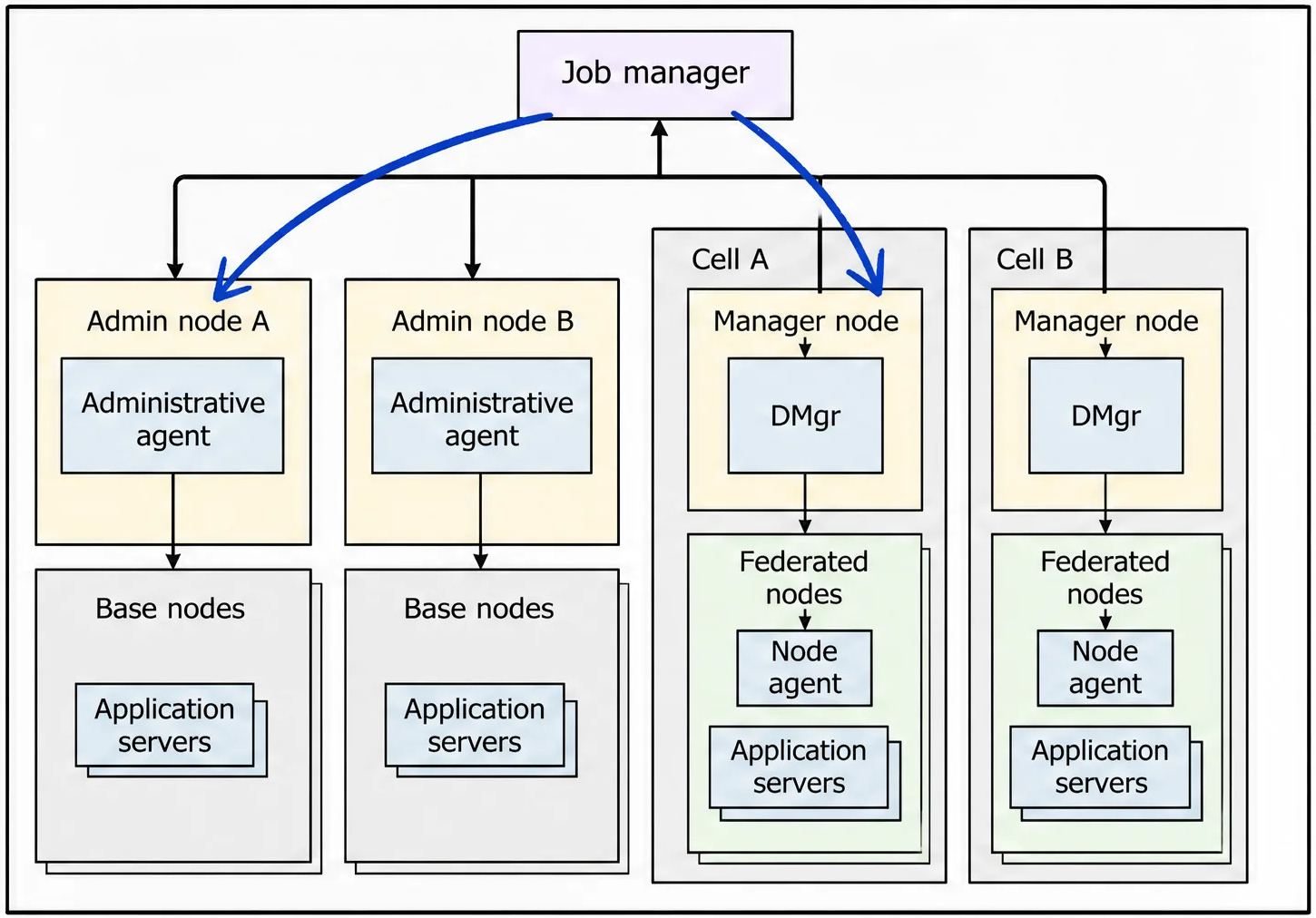
> “Registration depends on the right management relationship; a reachable JVM is not automatically a valid Job Manager target.”

Flexible-Management Topology

The Hub and Target Model



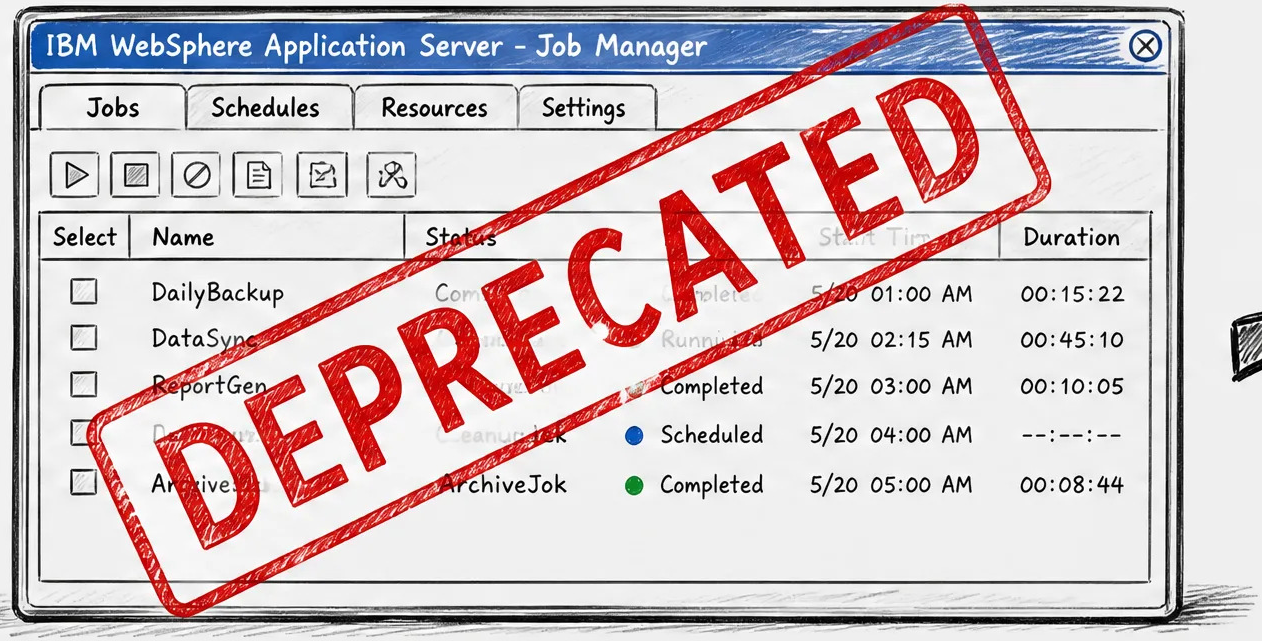
Automation must respect configuration boundaries even when a single console can submit work across them.



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Version Truth: Job Manager is Deprecated

Dr. Lee



→

UrbanCode Deploy & wsadmin

A Legacy-Operating Skill

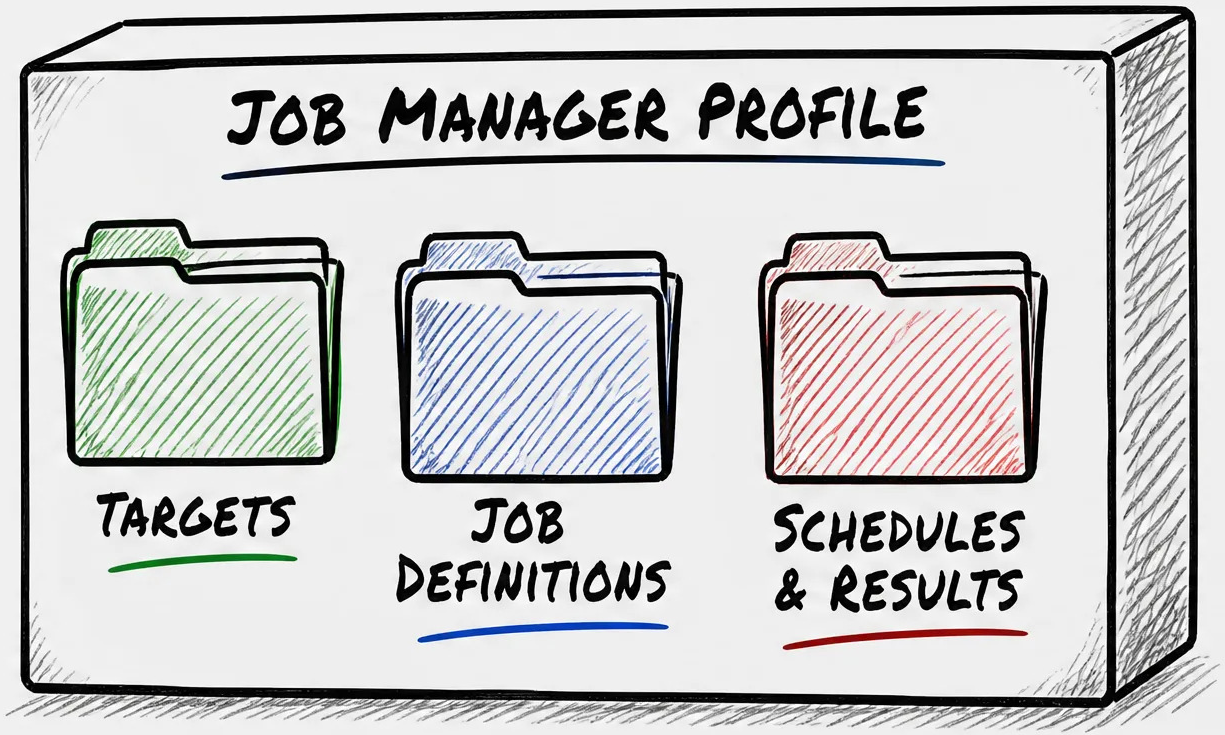
> "A legacy feature can be **operationally necessary** today without being **architecturally correct** for tomorrow."

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What a Job Manager Profile Owns

Dr. Lee

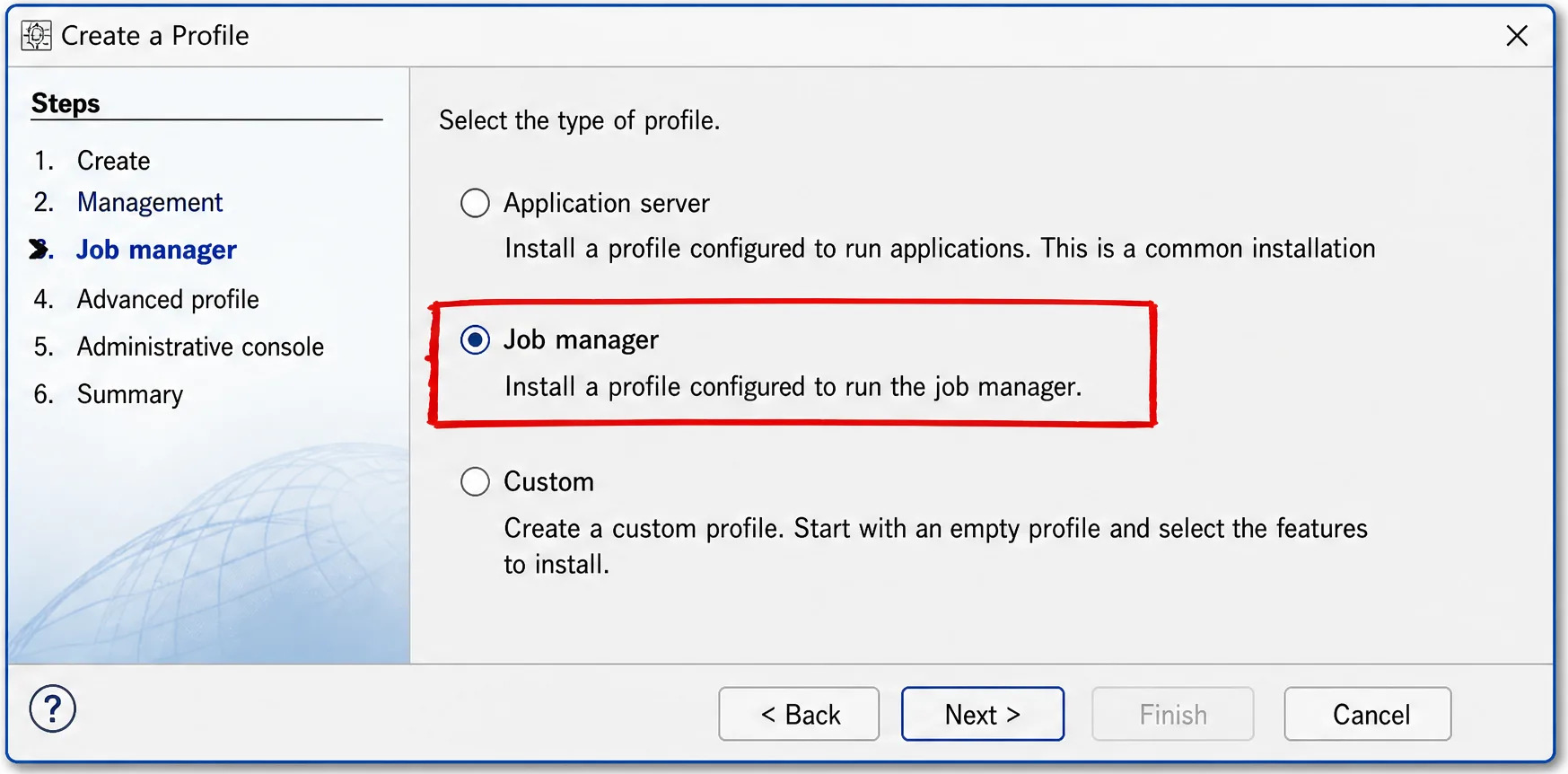
A Coordinator and a Queue



> "Submitting a job creates **Intent**;
> Target execution creates the **operational result**."

Create a Legacy Job Manager Profile

A Managed Runtime Boundary



The safest legacy deployment is one with a documented purpose, owner, and retirement path.

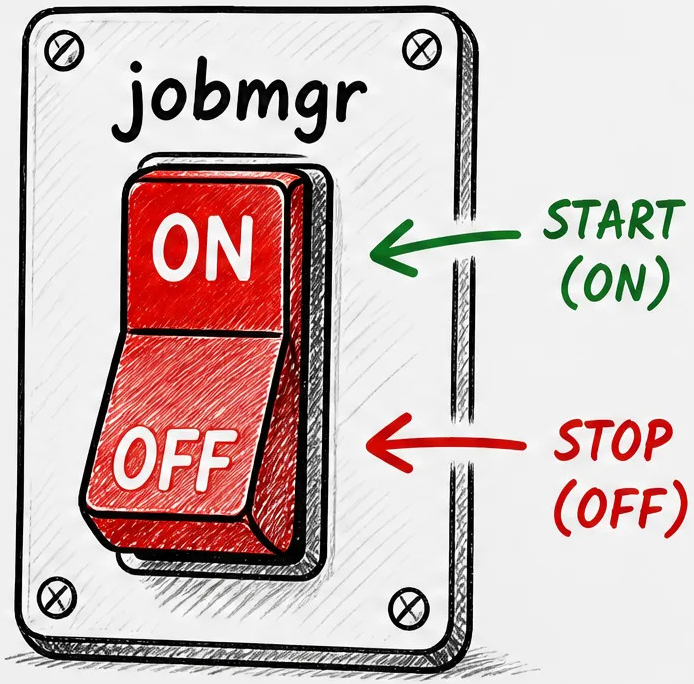
Start, Stop, and Configure the Job Manager

```
$ wsadmin> serverStatus

Server status for cell:  MyCell
Type                    Status
-------------------------------
AdminServer             RUNNING
NodeAgent               RUNNING
ApplicationServer1      RUNNING
ApplicationServer2      RUNNING
JobManager              RUNNING
DeploymentManager       RUNNING

The server status command completed.
```

Profile-Context Operations



> "A management queue **CANNOT** coordinate work while its own runtime is **UNAVAILABLE**."

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Preconditions Before Any Job Runs

Dr. Lee



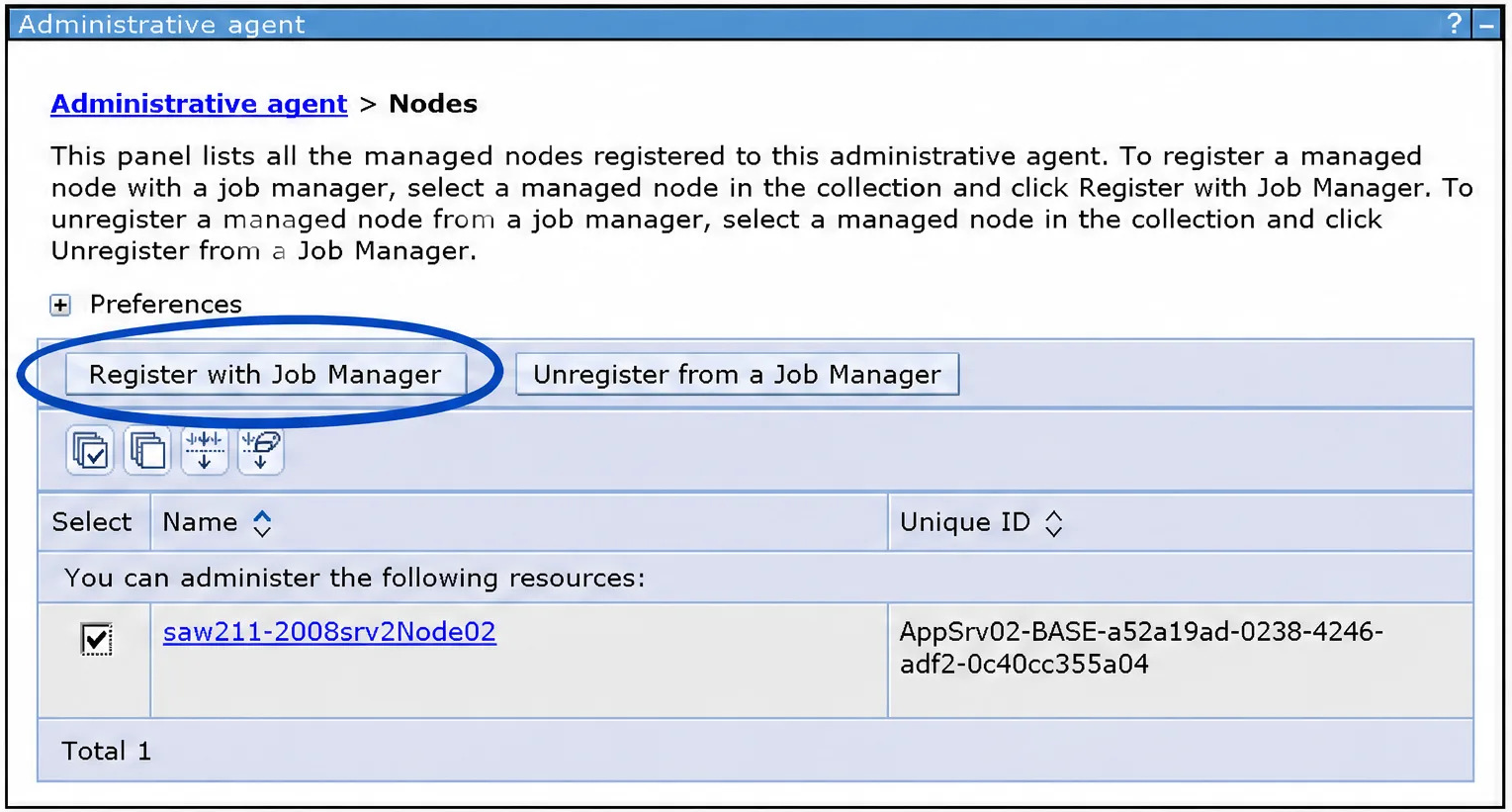
Target Readiness is the First Gate

1. Target Registered ☑
2. Target Reachable ☑
3. Target Started ☑
4. Eligible for Action ☑

> "Target Readiness is **the first verification gate**; Job submission is **NOT** the first action."

Register Administrative-Agent Targets

Creating a Trust Relationship





> The Job Manager **registers** the administrative agent's **managed node**, not a random **application-server process.**

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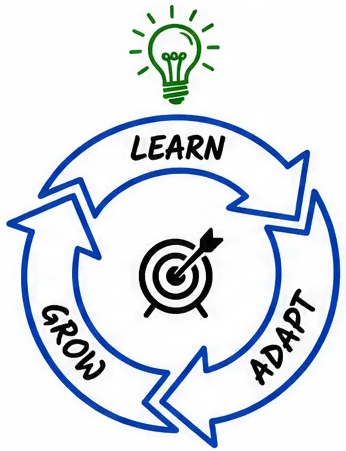
Register Deployment Managers and Hosts

DR. LEE



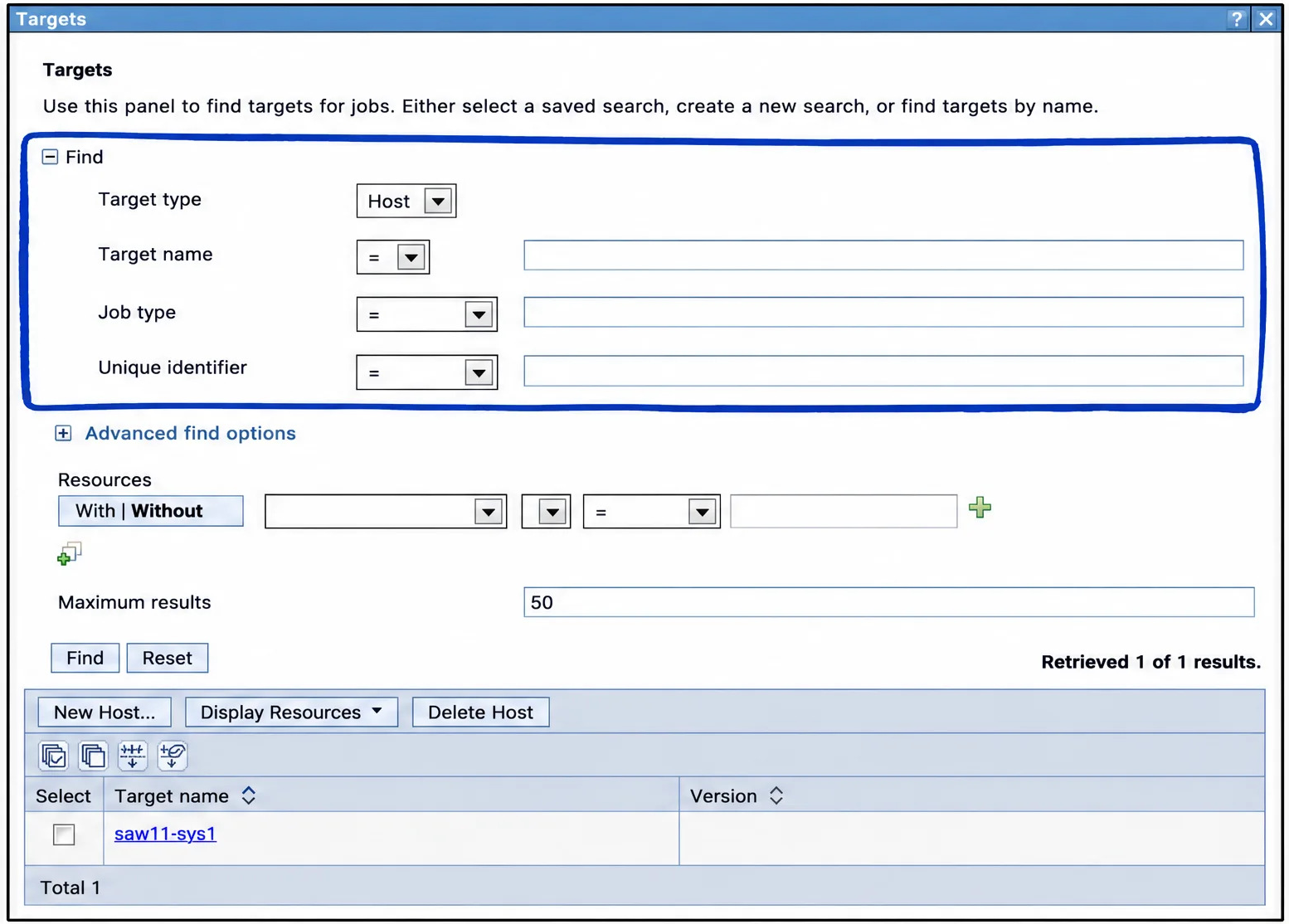
> CROSS-TARGET AUTOMATION SUCCEEDS ONLY WHEN AUTHENTICATION AND PARAMETERS ARE PORTABLE ACROSS EVERY SELECTED ENDPOINT.

Discover Targets, Resources, and Groups

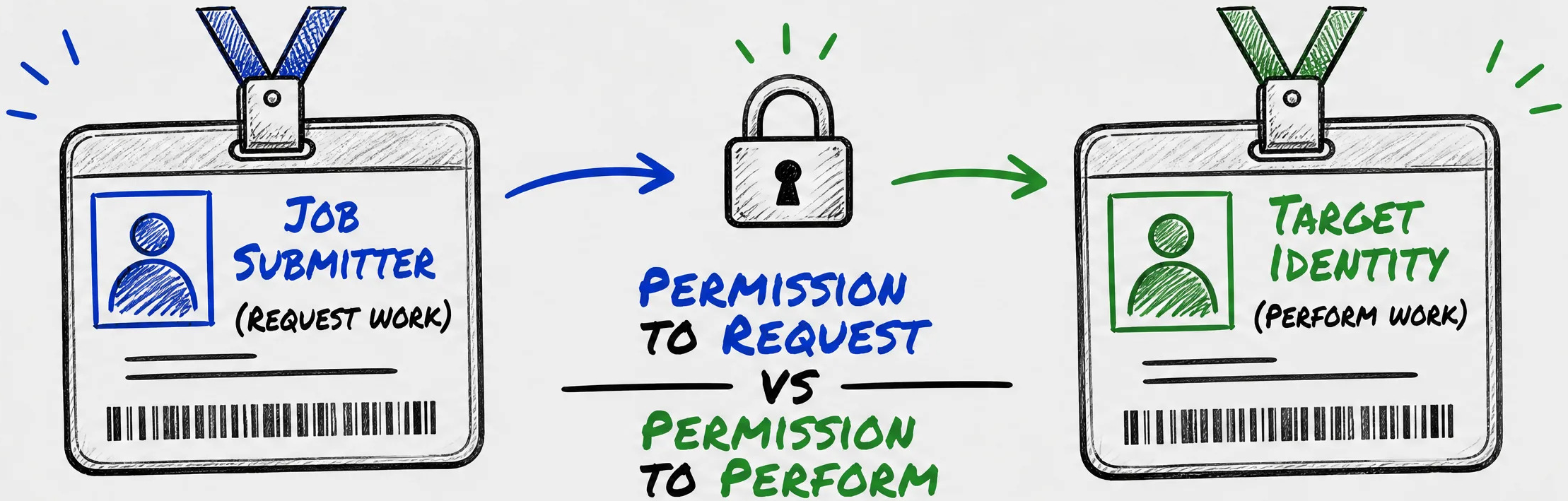


The Blast-Radius Inventory

Target discovery is pre-change validation, not an optional browsing step.



Authorization and Target Credentials

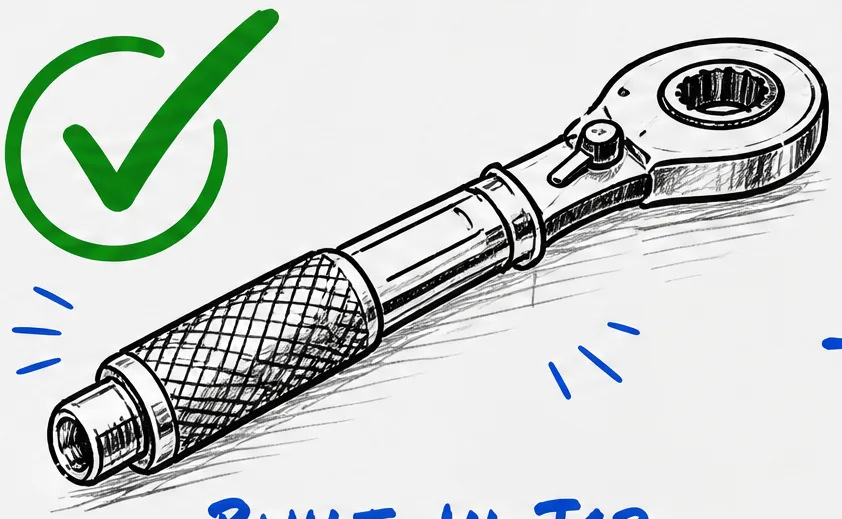


> “Automation Credentials are part of the Change Payload, not an invisible convenience.”

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Choose the Right Job Type

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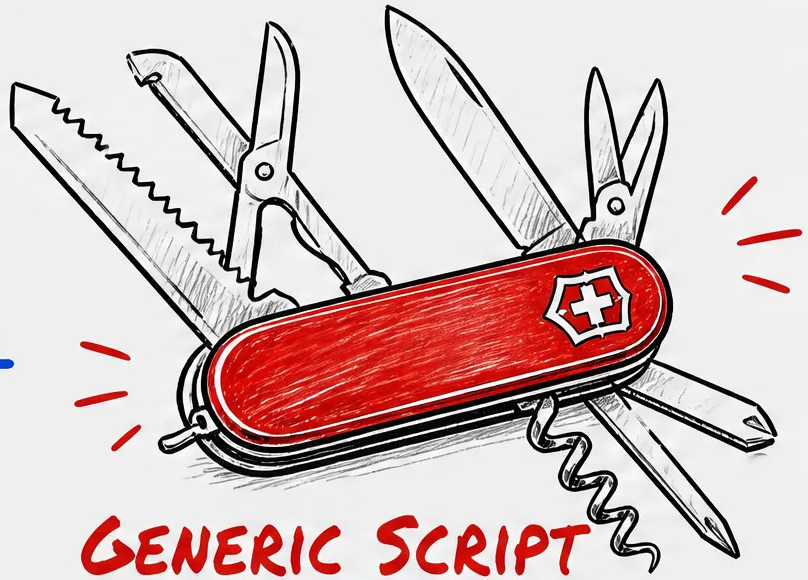


Built-in Job

(Precision Wrench)

✓ Specific, purpose-built, designed for one outcome. Minimal ambiguity.

The Contract for the Outcome



Generic Script

(Swiss-Army Knife)

✗ General-purpose, unbounded, adapted on the fly. Maximum ambiguity.

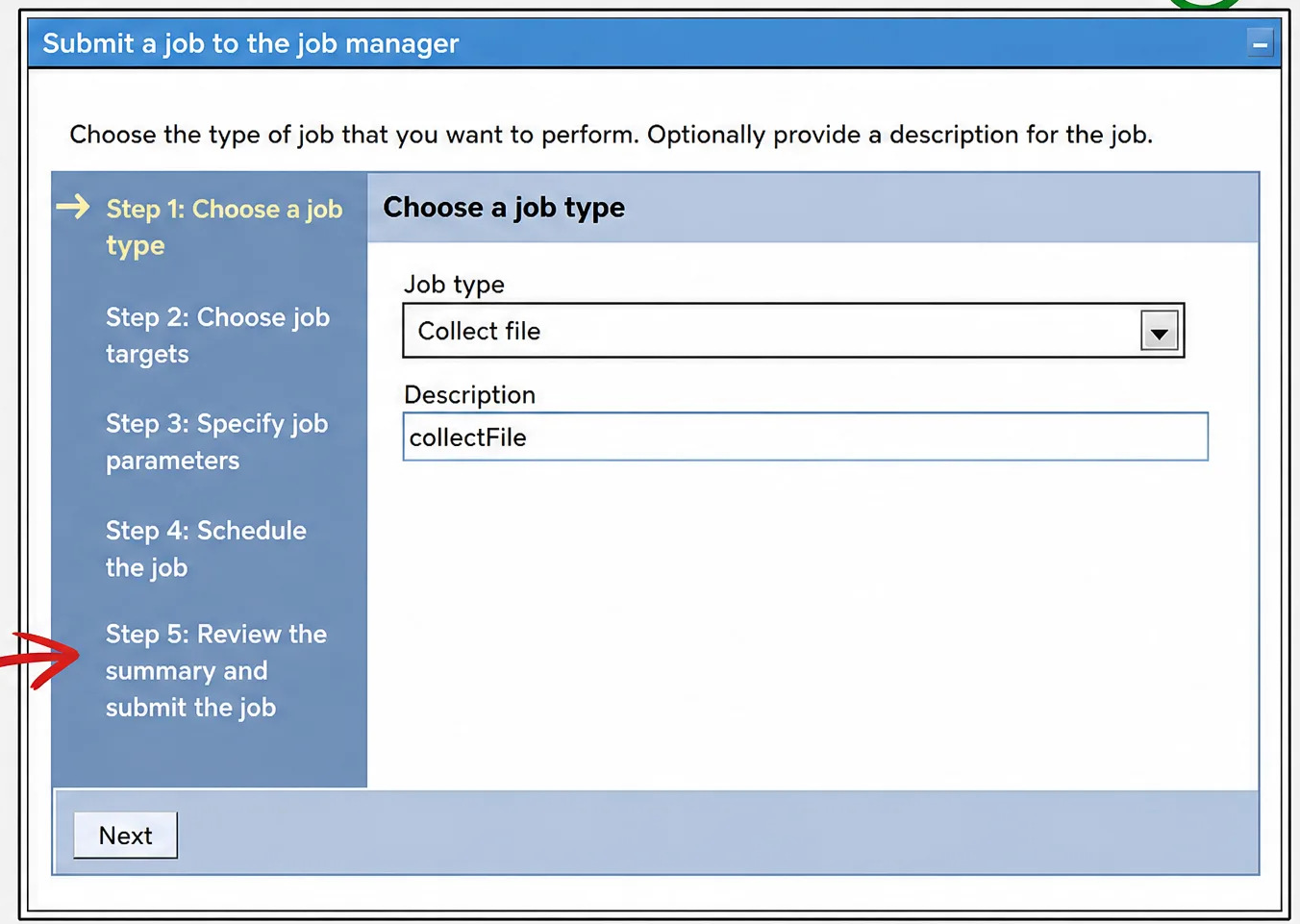
> "A specific **Job** reduces **ambiguity**; an **unbounded Script** expands it."

Submit a Job as a Controlled Change

More Than Just One Click

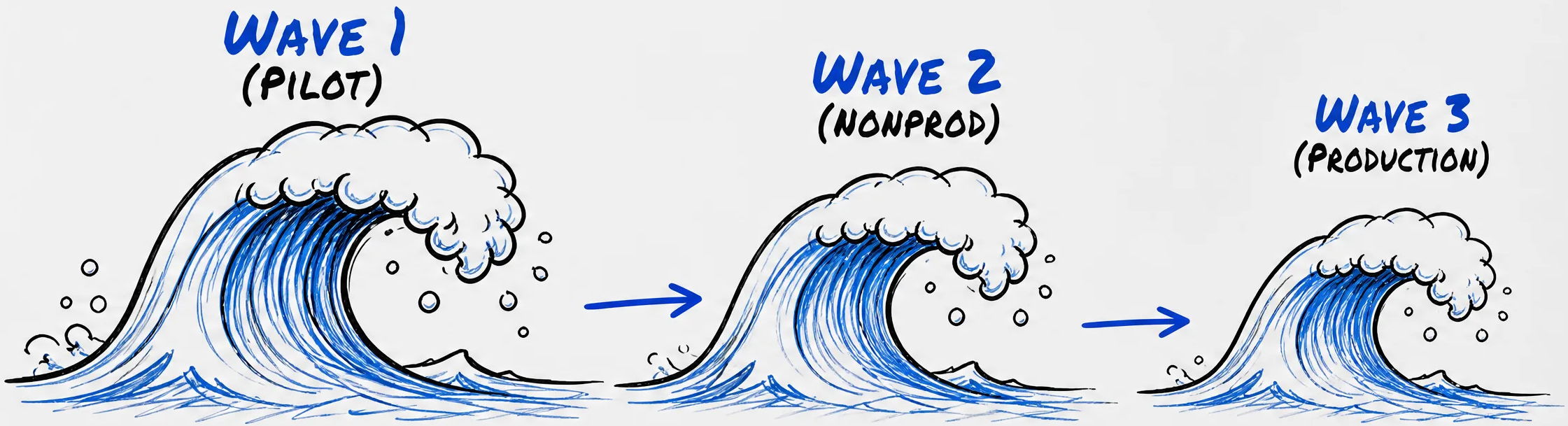
> "If a change cannot be explained clearly in the submission summary, it is not ready to fan out."





Target Groups Control Fanout

Dr. Lee



[A Scope, Not a Topology]

> “Fanout is a **deployment decision**:
> **small, verified waves** beat one large unexplained change.”

Schedule, Repeat, and Expire Work

Availability interval

Run once

- Run once
- Daily
- Weekly
- Monthly by day
- Monthly by date
- Yearly by day
- Yearly by date
- At connection

Controlling When Work is Valid

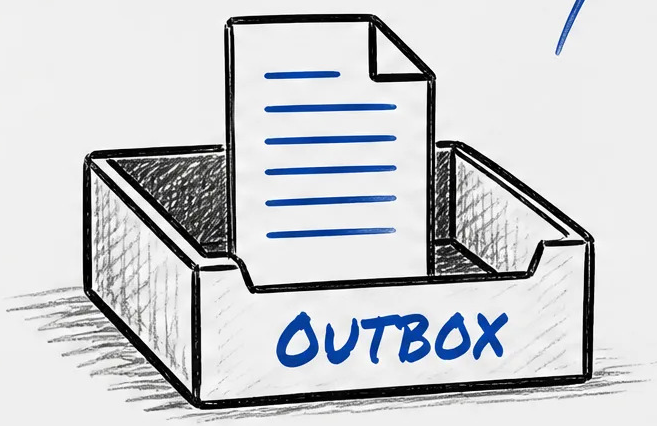


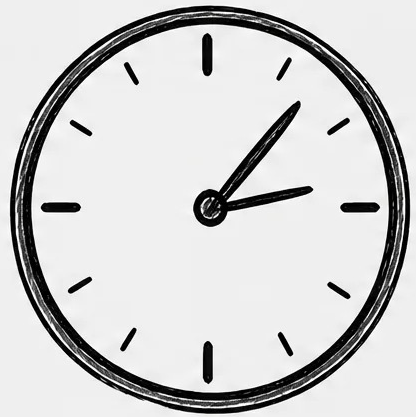
An unattended recurring job is still a change, so it needs the same ownership and stop conditions as a manual one.

Queueing and Polling Explain Delay

Outbox (Queue)

Submission accepted and placed in Outbox Queue.

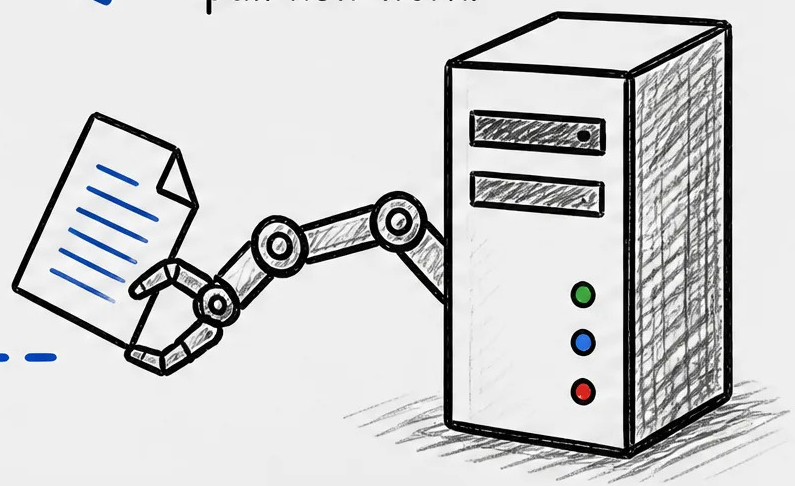




EVERY **30** SECONDS

Polling (Pull)

Target server reaches out every 30 seconds to pull new work.



Asynchronous by Design

> ⚠ Console submission is **not evidence** that target **execution** has begun.

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Read Job State and Target Status Separately

Job State

ACTIVE



The job request is available.

Coordinator View VS Endpoint View

Target Status (per endpoint)



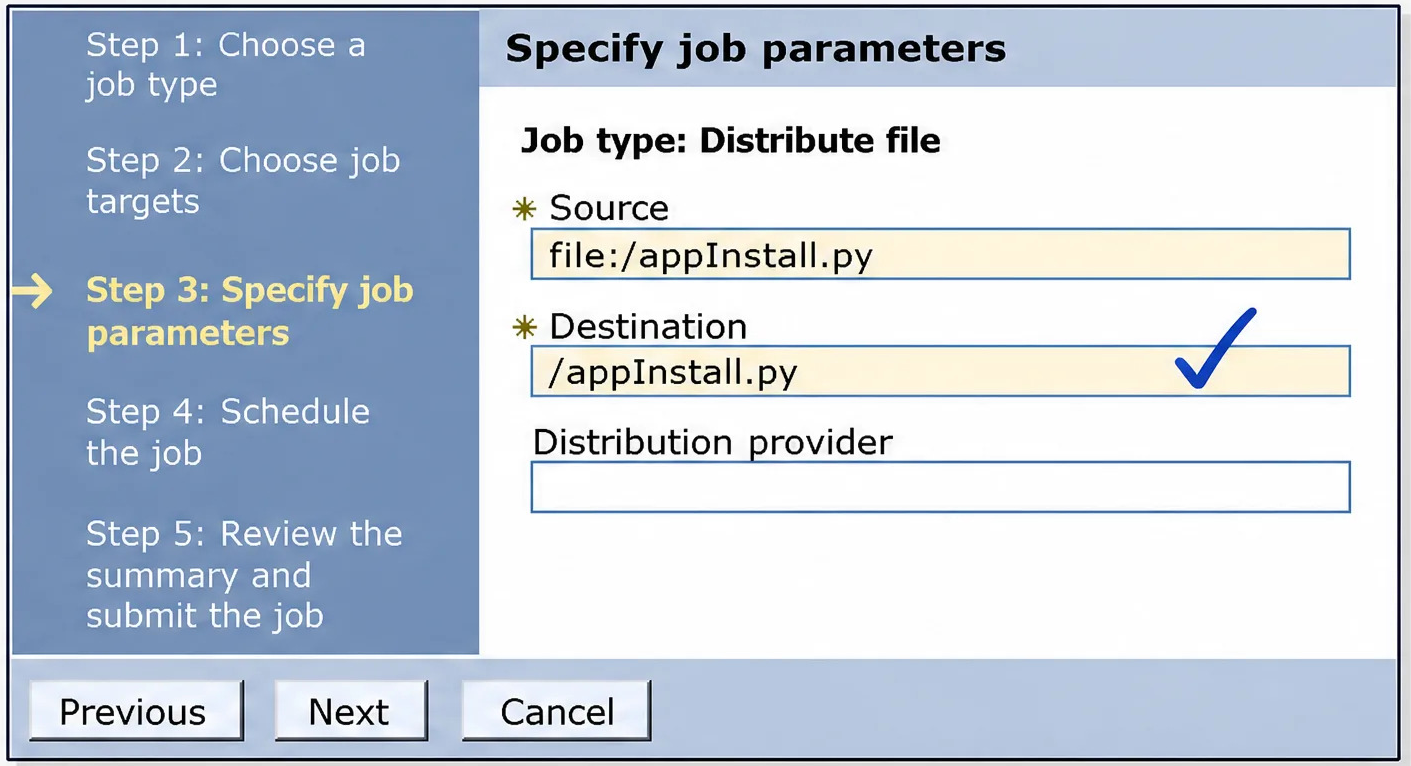
Target 1: Succeeded

Target 2: In Progress

Target 3: Failed

> Job State tells you whether the request is available;
> Target Status tells you what each endpoint actually did.

Stage Files Before Remote Work



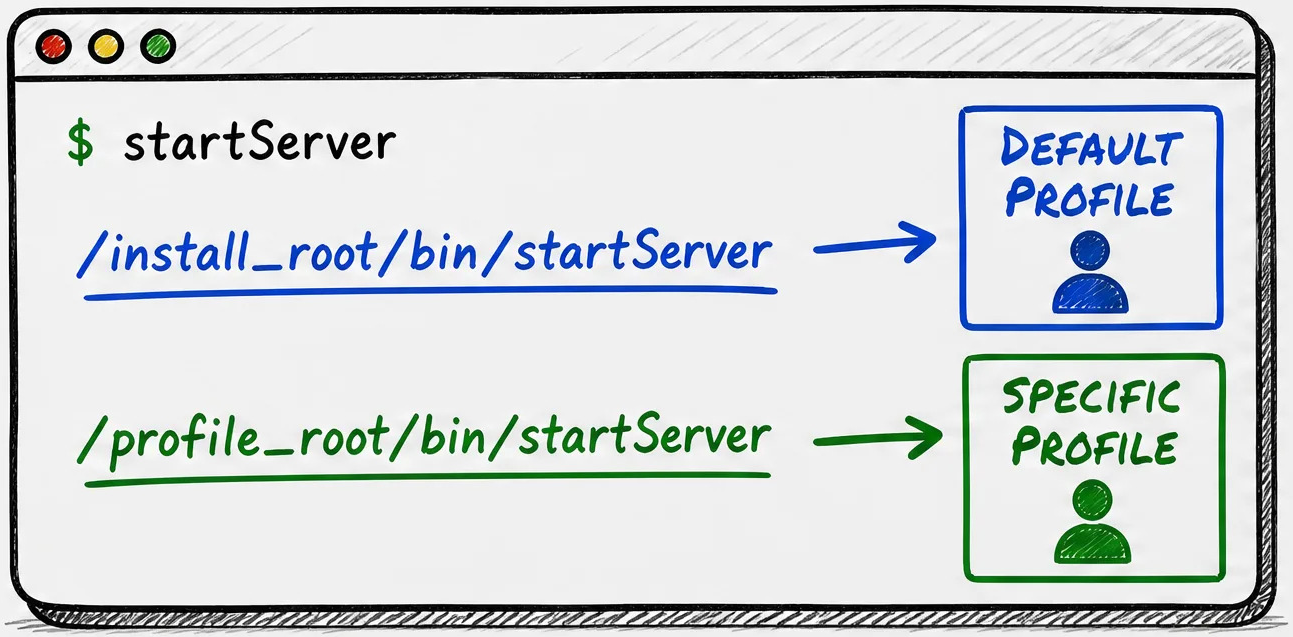


The Hidden Prerequisite



Remote execution is only repeatable when the **exact** required artifact is staged at the **exact** expected location.

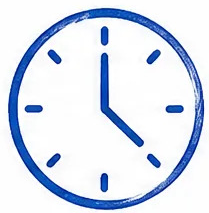
Command Location Determines Profile Context



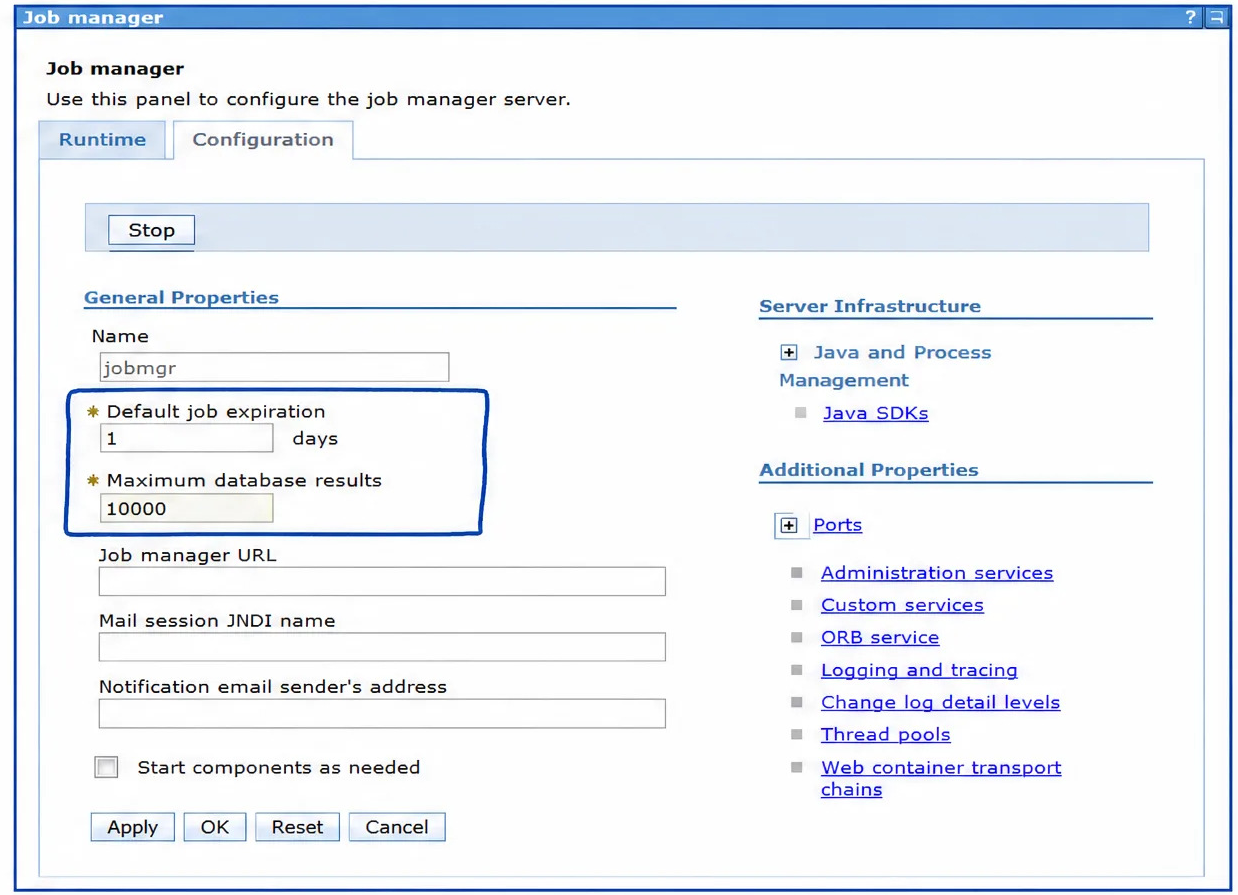
Context is **Not** a Shell Detail

Profile ambiguity is a **Change-Control** defect, **Not** a harmless shell detail.

Job Manager Configuration Is Queue Policy



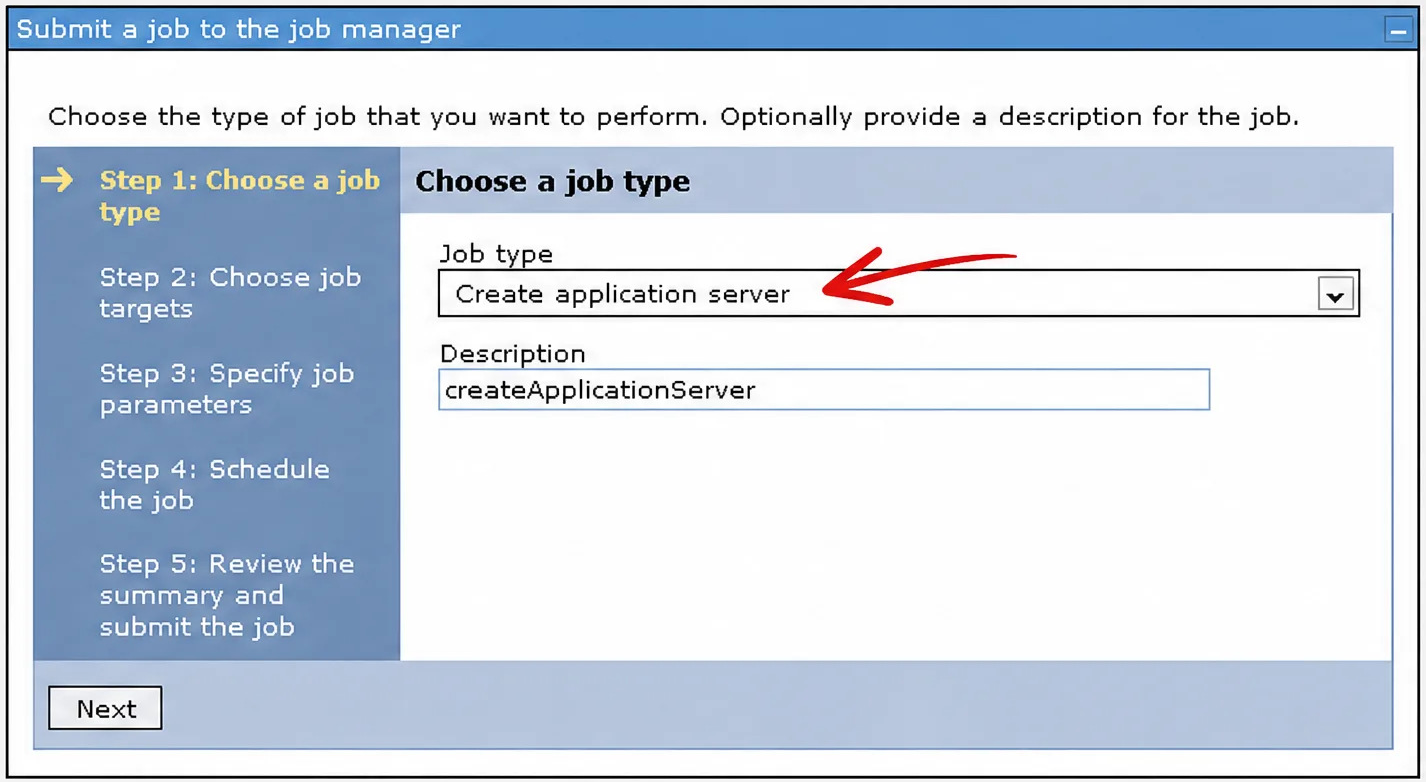
Time, Evidence, and Response





Memorable Statement: Queue configuration should be reviewed like any production policy because it controls *time*, *evidence*, and *operator response*.

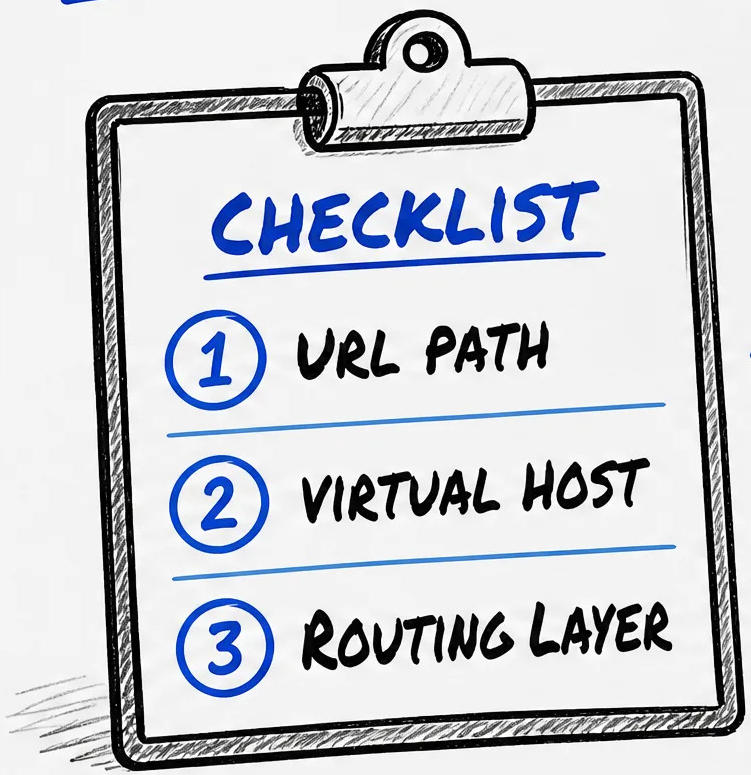
Create an Application Server Job



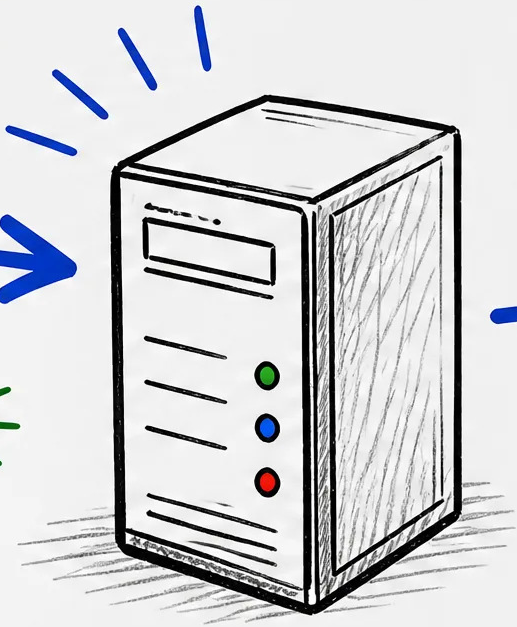
Multiplying Configuration Decisions

A successful create job proves the server object exists;
it does not prove the server is ready for application traffic.

Finish the Endpoint Contract



ALL THREE ALIGN



Reachable Service



> A running server is **not** a **reachable service** until the **URL path**, **virtual host**, and **routing layer** agree.

Verify Server Status and Job Outcome

The syntax of the **serverStatus** command is:

```
serverStatus.bat(sh) [options]
```

The options are shown Example 7-5.

Example 7-5 serverStatus options

```
Usage: serverStatus <server name | -all>
[-logfile <filename>]
[-replacelog]
[-trace]
[-username <username>]

[-password <password>]
[-profileName <profile>]
[-help]
```




Job Evidence vs Runtime Evidence



Automation is verified at the Target, not merely at the coordinating console.



Confirm Server health and Job outcomes directly on the target system.

Current Automation Pattern and Lab

Old Pattern



Deprecated

Controlled, Repeatable Administration

Modern Automation

wsadmin/Jython

For Scripts
- Versioned
- Reusable
- Reviewable

+

UrbanCode Deploy

For Deployments
- Controlled
- Auditable
- Repeatable

> Modern automation replaces a deprecated coordinator with versioned scripts, controlled deployment tooling, and explicit verification.